

Experiment 7

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7.1.2 Possible combinations from 3 digits

Algorithm

Step 1: Start the program.

Step 2: Read three space-separated digits as input (a, b, c).

Step 3: Store the three digits in a list or separate variables.

Step 4: Print the first combination by arranging digits as a, b, c.

Step 5: Print the second combination by arranging digits as a, c, b.

Step 6: Print the third combination by arranging digits as b, a, c.

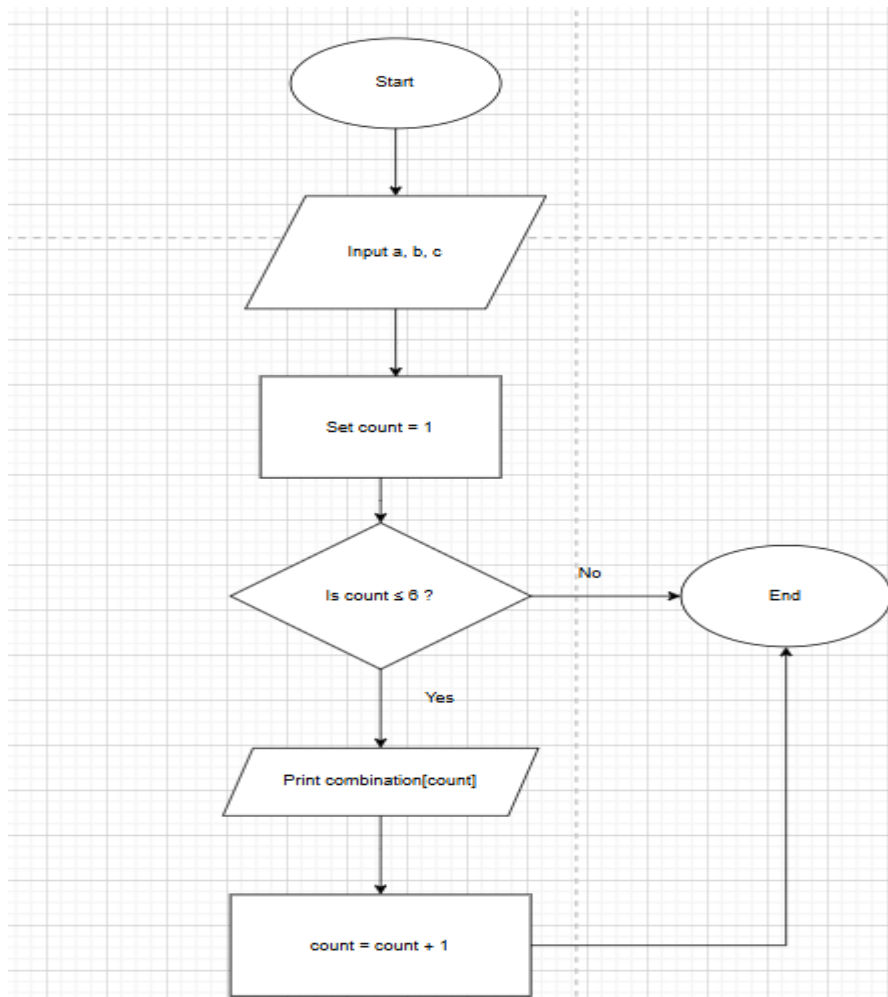
Step 7: Print the fourth combination by arranging digits as b, c, a.

Step 8: Print the fifth combination by arranging digits as c, a, b.

Step 9: Print the sixth combination by arranging digits as c, b, a.

Step 10: Stop the program.

Flowchart



Experiment 7

PYTHON CODE

```
a, b, c = input().split()

digits = [a, b, c]

for i in range(3):
    for j in range(3):
        for k in range(3):
            if i != j and j != k and i != k:
                print(digits[i] + digits[j] + digits[k])
```

EXECUTION

7.1.2. Possible Combinations from Three Digits

Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.

For three digits a , b , and c , there are 6 possible arrangements based on their indices: abc , acb , bac , bca , cab , cba .

Input Format:

- Single line contains three space-separated non-negative integers, a , b , and c , representing the three digits

Output Format:

- Print all possible 3-digit combinations
- Each combination should be printed as a 3-digit number on a separate line
- Print combinations as they appear (maintain leading zeros if first digit is 0)

Constraints:

- $0 \leq \text{each digit} \leq 9$
- Digits can be same or different

Sample Test Cases

```
1 a, b, c = input().split()
2
3 print(a + b + c)
4 print(a + c + b)
5 print(b + a + c)
6 print(b + c + a)
7 print(c + a + b)
8 print(c + b + a)
```

Test Results:

- Average time: 0.005 s (4.71 ms)
- Maximum time: 0.006 s (6.00 ms)
- 3 out of 3 shown test case(s) passed
- 4 out of 4 hidden test case(s) passed

Test case 1:

Expected output	Actual output
1 2 3	1 2 3
123	123
132	132
213	213
231	231
312	312

Terminal | Test cases

Prev | Reset | Submit | Next